OXIDATION REDUCTION TITRATIONS-II

Week 8 Lecture

PCC/PCD 105

Standard Iodine solution, I_3^-

- Properties
- weak oxidant… only selective for strong reducing agents
- brown , but not self ind. <u>Except</u> in case of karl-Fischer

$I_2(s) + \quad I^- \leftrightarrow I_3^-$

- Stability of standard solution…. 2ry standard

  $I^- + O_2(g) + 4H^+ \rightarrow I_2 + 2H_2O$ … increase conc.

  I_2 is volatile … decrease conc.

- Standardization

  $H_3AsO_3 + I_2 + H_2O \leftrightarrow H_3AsO_4 + 2I^- + 2H^+$, <u>($NaHCO_3$)</u>

$I_2 + 2S_2O_3^{2-} \leftrightarrow S_4O_6^{2-} + 2I^-$

Reduction half reaction

- $I_2 + 2e \leftrightarrow 2I^-$ $E^0 = 0.536$ V

- **OR**

- $I_3^- + 2e \leftrightarrow 3I^-$
- *Eq. wt = M wt/ 2 for 1 mole iodine.*

Titrations involving Iodine

❑ Iodimetric titration:

- Direct
- $H_3AsO_3 + I_2 + H_2O \leftrightarrow H_3AsO_4 + I^- + 2H^+$ ($NaHCO_3$, starch)
- Ep detection

  Starch is used as indicator It is added at the beginning of the titration. The first drop excess of I_3^- (I_2) after the equivalence point causes the solution to turn blue.

N.B. starch is not a true redox indicator

- Conditions:

  Titrations occur in neutral or weakly alkaline solutions

  $$AsO_4^{3-} + 2H^+ + 2e \leftrightarrow AsO_3^{3-} + H_2O$$

  $$E = E^o - \frac{0.059}{2}\log\frac{[AsO_3^{3-}]}{[AsO_4^{3-}][H^+]^2}$$

  1- $[H^+] \downarrow$ E $\downarrow$ of the sample

  2- Equilibrium shift to the right of the overall reaction.

$$H_3AsO_3 + I_2 + H_2O \leftrightarrow H_3AsO_4 + I^- + 2H^+$$ ($NaHCO_3$, starch)

- **How to force iodine-analyte reaction toward completion**
- $Fe^{2+} + I_2 \leftrightarrow Fe^{3+} + 2I^-$
- **pH effect**

  $I_2 + 2e \leftrightarrow 2I^-$ $E^0 = 0.536$ V

  pH has no effect on electrode potential.
- **Complexing agent**

  Fe^{3+}/ Fe^{2+} system,

  $Fe^{3+} + e^- \leftrightarrow Fe^{2+}$ $E^0 = 0.77$ V

  $I_3^- + 2e \leftrightarrow 3I^-$ $E^0 = 0.536$ V
- Addition of complexing agents as F^- or EDTA that form stable cpx with Fe^{3+} → shift to lower E (E < 0.53 V) that allows oxidation of Fe^{2+} with I_2

Sources of error in iodimetry

- **Due to I_2**

  Lack of stability (volatility, air oxidation of iodide)

- **Due to starch**

  Decomposition products,

e.g. -glucose may be oxidized with iodine+(Ve error)

- other products give irreversible reddish color with I_2.

To Avoid, add preservative like boric acid or formamide.

Applications in Iodimetry

- **By Direct Titrations**

Determination of $Na_2S_2O_3$

$$\mathbf{2S_2O_3^{2-} + I_2 \leftrightarrow S_4O_6^{2-} + 2I^-}$$

tetrathionate

Determination of As^{3+}

$$\mathbf{H_3AsO_3 + I_2 + H_2O \leftrightarrow H_3AsO_4 + 2I^- + 2H^+}$$

- Starch ind, added at the beginning of titration
- The solution is neutral or slightly alkaline , use $NaHCO_3$

❑ Iodometric titration:

- Indirect
- For analyzing oxidizing agents
- Add xss I^- to a solution of analyte. I_2 is produced in an amount equivalent to the oxidizing agent. Liberated I_2 is tit ≠ st $Na_2S_2O_3$

$$Cr_2O_7^{2-} + 6\ I^-(xss) + H^+ \leftrightarrow Cr^{3+} + 3I_2 + H_2O$$

$$I_2 + 2S_2O_3^{2-} \leftrightarrow I^- + S_4O_6^{2-}$$

$$Cr_2O_7^{2-} \equiv 3I_2 \equiv 6S_2O_3^{2-}$$
$$\equiv 6\ e$$

$Na_2S_2O_3$ is a universal titrant for I_2 in neutral or acidic solutions

- Ep detection

  Starch near E.P.... (if added in the beginning will form irreversible blue color)

  $CHCl_3$ or CCl_4

Titrations occur in acidic solutions

$MnO_4^- + 8H^+ + 5e \leftrightarrow Mn^{2+} + 4H_2O$

$$E = E^o - \frac{0.059}{5} \log \frac{[Mn^{2+}]}{[MnO_4^-][H^+]^8}$$

$[H^+] \uparrow$ $E \uparrow$

(titration should be performed rapidly to minimize air oxidation of iodide)

- **How to force oxidation of iodide by analyte toward completion**

1-pH

↑ $[H^+]$ promotes oxidizing agent-iodide reaction (↑E of oxidizing agent couple)

2- Precipitating agent

$Cu^{2+} + e^- \leftrightarrow Cu^+$

$E^0\ Cu^{2+}/Cu^+ = +0.34\ V$ while $E^0\ I_2/I^- = +0.536\ V$

adding SCN^- OR I^- reagent to ppt Cu^+ --- $[Cu^+]$ ↓ → ↑E of Cu^{2+}/Cu^+

$$E = E^o - \frac{0.059}{1} \log \frac{[Cu^+]}{[Cu^{2+}]}$$

$[Cu^+]$ ↓ E ↑

Function of iodide

(a) reduce Cu^{2+} to Cu^{+}

(b) precipitate Cu^{+} ion

$$2Cu^{2+} + 4I^{-} \longrightarrow Cu_2I_2 \downarrow + I_2$$

But the precipitated Cu_2I_2 would adsorb I_2 on its surface, so we add $(NH_4)SCN$ to precipitate the more insoluble $Cu_2(SCN)_2$ and set free the adsorbed I_2.

$$Cu_2I_2 + 2SCN^{-} \longrightarrow Cu_2(SCN)_2 + 2I^{-}$$

Sources of error in iodometry

- Decomposition of thiosulfate solution.
- Premature addition of starch and
- Starch decomposition.
- **<u>Due to I_2</u>**

  Lack of stability (volatility, air oxidation of iodide)

Iodometric

Determination of oxidizing agents

$$2MnO_4^- + 10I^- + 16H^+ \leftrightarrow 2Mn^{2+} + 5I_2 + 8H_2O$$

$$Cr_2O_7^{2-} + 6I^- + 14H^+ \leftrightarrow 2Cr^{3+} + 3I_2 + 7H_2O$$

$$Cl_2 + 2I^- \leftrightarrow I_2 + 2Cl^-$$

$$BrO_3^- + 6I^- + 6H^+ \leftrightarrow Br^- + 3I_2 + 3H_2O$$

$$IO_3^- + 5I^- + 6H^+ \leftrightarrow 3I_2 + 3H_2O$$

$$2Fe^{3+} + 2I^- \leftrightarrow 2Fe^{2+} + I_2$$

$$H_3AsO_4 + 2I^- + 2H^+ \leftrightarrow H_3AsO_3 + I_2 + H_2O$$

Titrate the liberated **I_2** ≠ st $\underline{Na_2S_2O_3}$, CCl_4 ind

$$2S_2O_3^{2-} + I_2 \leftrightarrow S_4O_6^{2-} + 2I^-$$

Titrations involving Iodine

Iodimetry

- Weakly alkaline or neutral solution
- **If pH is too alkaline**

I_2 will disproportionate to hypoiodide and iodide

$I_2 + 2OH^- \leftrightarrow IO^- + I^- + H_2O$

OI^- is a stronger oxidant than I_2

Iodometry

- Acidic solution
- **Why titration solution kept NOT strongly acidic**

  Starch ind. tends to hydrolyze in strong acid.

  I^- tends to be oxidized by dissolved oxygen. In strongly acidic medium

  $4I^- + O_2(g) + 4H^+ \rightarrow 2I_2 + 2H_2O$

karl Fischer reagent for determination of water

- Reagent

Iodine, sulfur dioxide dissolved in pyridine (I_2, SO_2 present as complexes *to* prevent their volatilization) and xss. methanol

C_5H_5N $C_5H_5N.I_2$ $C_5H_5N.SO_2$
MeOH

- Application

Determination of water in organic liquids and solids which are soluble in methanol and can be analyzed by direct titn (acids, alcohols, esters, anhydrides…)

- $C_5H_5N.I_2 + C_5H_5N.SO_2 + C_5H_5N + H_2O \leftrightarrow 2C_5H_5N.HI + C_5H_5N.SO_3$

$$I_2 + SO_2 + H_2O \leftrightarrow 2HI + SO_3$$

- Method: Direct titration
- E.p.: potentiometry

 Color change from yellow (reaction products) to brown (xss reagent, pyridine iodine cpx)

- Stoichiometry: 1 mole of I_2, 1 mole of SO_2, for each mole of H_2O.

$I_2 \equiv H_2O$

Role of methanol in Karl Fisher Reagent

- $C_5H_5N.SO_3 + H_2O \leftrightarrow C_5H_5NH.SO_4H$
- $C_5H_5N.SO_3 + CH_3OH \leftrightarrow C_5H_5NH.SO_4CH_3$

- **Prevent reaction of product "sulfur trioxide" with water sample eliminating the susceptible –ve error**


Metrohm
800 Dosino
10 mL
Riedel-de Haën
HYDRANAL-Composite 5
HYDRANAL-Methanol Rapid
803 Ti Stand
Titrando